

SAANVI REDDY BARADI

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SUMMARY

AI Engineer who builds generative AI systems end to end, from RAG pipelines and fine-tuned LLMs to agentic workflows and NL-to-SQL engines, and takes them all the way into production. Comfortable across TensorFlow, PyTorch, LangChain, LangGraph, QLoRA/PEFT, Vertex AI (Gemini), Groq, Pinecone, and pgvector, with a particular focus on LLM evaluation, hallucination mitigation, and shipping AI systems that hold up under real use.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain (Chains, Agents, LCEL), Agentic Workflows (LangGraph, ReAct), QLoRA/PEFT Fine-Tuning (Unsloth), Vertex AI (Gemini), LangSmith, OpenAI API, Groq (Llama 3.3), RAG Pipelines, Prompt Engineering, LLM Evaluation (BLEU, ROUGE, Ragas), Hallucination Detection & Mitigation

AI/ML Engineering: TensorFlow, PyTorch, Scikit-learn, XGBoost, SHAP, SMOTE, NLP (spaCy, NLTK, TF-IDF, NER), Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Data Engineering: Python, Pandas, ETL Pipeline Design, RapidFuzz (Fuzzy Matching), Data Validation & Quality Assurance, Schema Validation, REST API Integration (U.S. Census Bureau API)

Vector Databases & Embeddings: Pinecone, ChromaDB, PostgreSQL (pgvector), Hugging Face Embeddings (all-MiniLM-L6-v2), Vertex AI Embeddings, Semantic Search, Similarity Retrieval

Software & APIs: Python, FastAPI, RESTful API Design, Pydantic, SQL, pytest, Git/GitHub, CI/CD (GitHub Actions), Docker, Error Handling, Rate Limiting

Cloud & MLOps: GCP (Vertex AI, Cloud Run, Cloud Logging), AWS EKS, Azure AKS, Kubernetes, Helm, Terraform, Docker & Compose, Streamlit, Model Serving, Containerization

BI & Visualization: Tableau, Power BI (DAX, PL-300 Certified), Matplotlib, Seaborn

EXPERIENCE

AI & Data Analyst – Compass Group / University of South Florida

Jul 2026 – Present

- Work with research and domain stakeholders to turn AI and data problem statements into technical requirements, analytical workflows, and measurable evaluation criteria, then validate results with Python, SQL, and statistical analysis.
- Prototype RAG-based workflows spanning document processing, chunking, embeddings, vector retrieval, and LLM response generation, and design evaluation experiments that measure relevance, factual consistency, and retrieval quality.
- Communicate analytical findings, AI evaluation results, and limitations to stakeholders, and fold that feedback into the next round of experiments.

Project Assistant – AI Systems & RAG – University of South Florida

Feb 2025 – May 2026

- Built Python-based data pipelines for AI applications, handling ingestion, preprocessing, feature engineering, schema validation, and metadata generation, which cut manual data-processing effort by roughly 70%.
- Wrote reusable data-quality and validation modules for AI-ready datasets, adding anomaly detection, duplicate detection, and exception handling that made semantic search, machine learning, and RAG workflows more reliable.
- Put together end-to-end RAG workflows, from document preprocessing and embedding generation to vector search and LLM-based response generation, and used RapidFuzz-based entity resolution to cut country-name mismatches by about 90% across 180+ variations; automated testing and validation through Git-based CI/CD.
- Integrated the U.S. Census Bureau ACS 5-Year API with FIPS-based logic to pull demographic and income data at the ZIP, county, and state level for downstream AI and analytics use.

Data Scientist – Accenture

Jun 2023 – May 2024

- Built Python and SQL pipelines processing 5M+ records across cognitive and socioeconomic datasets, handling ingestion, transformation, validation, and feature preparation for machine learning workflows.
- Developed and evaluated machine learning models in Python and scikit-learn, covering feature engineering, model validation, and performance evaluation for predictive analytics use cases.
- Designed SQL-based data models and automated reporting pipelines that supported 8+ downstream analytics and machine learning use cases, and built Tableau/Power BI dashboards that cut manual reporting effort by 30%.

Research Assistant – Malla Reddy Engineering College for Women

Jun 2021 – Jun 2023

- Built an end-to-end NLP and generative AI system to analyze 18,617 OSHA construction accident reports, combining text preprocessing, TF-IDF, NER, and SMOTE with supervised ML classification to predict accident risk at 96% accuracy and 0.9867 AUC-ROC.
- Brought Google Gemini into the pipeline, writing the prompt construction and response-processing logic that turned model predictions and extracted entities into structured, context-aware safety recommendations.
- Deployed the system as a containerized Streamlit and Docker app for real-time risk prediction and Gemini-powered recommendations, and built additional XGBoost/Random Forest classification pipelines with cross-validation that reached 89%+ accuracy across three classification tasks.

PROJECTS

RAG-Based Document Q&A System

LangChain, FastAPI, ChromaDB, pgvector, Vertex AI, Groq, Streamlit, Pytest

- Built and deployed a production RAG system end to end: PDF ingestion, Hugging Face embeddings (all-MiniLM-L6-v2), ChromaDB storage, and Groq LLM (Llama 3.3) answer generation, live on Hugging Face Spaces at under 500ms latency.
- Architected a dual-backend design, pgvector (PostgreSQL) for enterprise deployments and ChromaDB for lightweight ones, and used an LLM evaluation framework (ROUGE, fact consistency, retrieval quality) to track down and fix hallucination failures.

AI-Powered SQL Analytics Assistant

LangGraph, Groq (Llama 3.3 70B), Python, SQLite, Streamlit, SQL

- Built and deployed an agentic NL-to-SQL pipeline using LangGraph ReAct agents and tool orchestration, powered by Llama 3.3 70B via Groq, with conversational memory for multi-turn, session-aware queries.
- Added schema-aware prompting with live metadata injection and SQL guardrails (SELECT-only enforcement, JOIN validation, LIMIT constraints), reaching 95%+ query accuracy while blocking unsafe execution.

Multi-Cloud Fine-Tuned LLM Serving Pipeline

PEFT/QLoRA, Pinecone, FastAPI, Docker, Kubernetes, Terraform, GitHub Actions

- Fine-tuned Mistral-7B with QLoRA (4-bit, LoRA adapters via Unsloth) on a 24K-example support dataset, bringing training loss down from 0.79 to 0.63, and built a Pinecone-backed RAG layer for grounded retrieval.
- Set up an automated Ragas evaluation gate inside a GitHub Actions CI/CD workflow, containerized the serving API with FastAPI and Docker, and wrote cloud-agnostic Helm charts and Terraform IaC validated for AWS EKS and Azure AKS.

AI-Ready Real-Time Analytics Lakehouse

Kafka, Databricks, Snowflake, dbt, Airflow, LangChain, Groq

- Built a streaming pipeline from Kafka through Databricks PySpark Structured Streaming into Snowflake, with a clustered warehouse, materialized views, and a tested three-layer dbt model orchestrated by Airflow and gated by GitHub Actions CI/CD.
- Added a LangChain and Groq layer that lets someone ask the warehouse a question in plain English and get an accurate answer back, tying the data engineering and LLM work together.

EDUCATION

University of South Florida – Master of Science, Artificial Intelligence and Business Analytics, Tampa, FL

Aug 2024 – May 2026

Malla Reddy Engineering College for Women – Bachelor of Technology, Data Science, Hyderabad, India

Jun 2020 – Apr 2024

CERTIFICATIONS

Data Science Fundamentals (Python/SQL) – IBM | Introduction to Data Science – IBM | Power BI Data Analyst (PL-300) | Programming Essentials – Cisco | Python – HackerRank